Strand 3: Geometry and Measurement

Sub-strand 2: Measurement

Lesson 3 (Preamble): Perimeter and Area — Squares, Rectangles and Rectilinear Shapes

1. Curriculum Standard and Overview

- Strand: 3 — Geometry and Measurement
- Sub-strand: 2 — Measurement
- Content Standards: B7.3.2.1 foundations (perimeter, drawing figures on a square grid — B7.3.2.1.1, B7.3.2.1.3), building towards B8.3.2.1 area work.
- Indicators served: B7.3.2.1.3 (draw rectangles and triangles with given dimensions on a square grid), plus the perimeter/area foundations every later area lesson stands on.
- Why this lesson first: Before we can learn area of polygons or parts of a circle, you must be able to (a) find the length of the boundary of a shape, and (b) understand what "area" means as a count of unit squares. These two ideas are today's whole lesson.

Prerequisites already delivered: Lessons 1–13 (Number, Algebra, and the first Geometry lessons: Polygons & Angle Properties, Congruence & Similarity, Angles/Parallel Lines/Trigonometry, Circles/Scale quick pass, Vectors). We now restart measurement properly and slowly, because area work was only touched lightly in the mock-driven quick lessons.

2. Preamble — Two Different Questions About One Shape

Look at any flat shape (a rectangle, a football pitch, your exercise book). Mathematicians ask two different questions about it:

1. How long is the boundary? — the perimeter.
2. How much surface is inside the boundary? — the area.

- Perimeter is a length. You answer in cm, m, km. Imagine walking slowly around the edge of the shape and counting your steps. The total steps = perimeter.
- Area is a surface. You answer in square units: cm^2, m^2, km^2. Imagine covering the shape with little 1 cm × 1 cm squares and counting them.

Memory hook: peRimeter → aR ound the edge. Area → All the surface inside.

3. Core Concepts

3.1 Perimeter of a rectangle and a square

A rectangle has two lengths (l, twice) and two widths (w, twice), so:

$$P = l + w + l + w = 2(l + w)$$

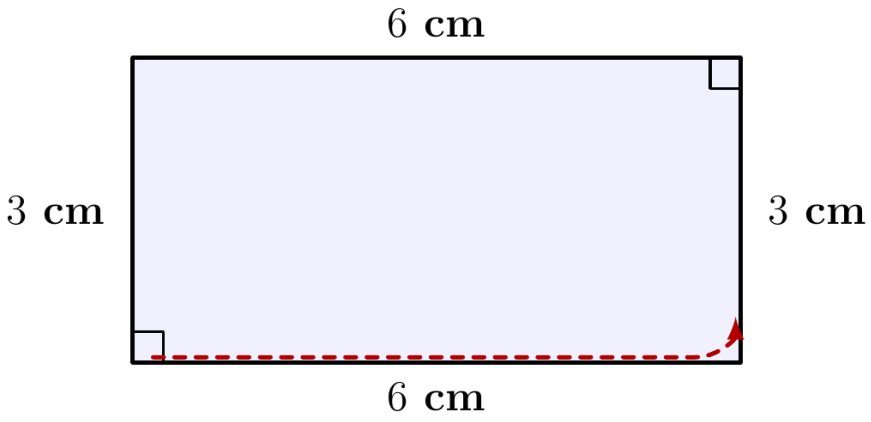


Figure 2 walks you round the fence once: the two lengths and two widths are the whole journey.

A square has four equal sides s, so:

$$P = 4s$$

3.2 Area of a rectangle and a square

A rectangle of length l and width w contains l rows of w squares, i.e. l × w squares:

$$A = l \times w \quad \text{(in unit squares, e.g. cm}^2\text{)}$$

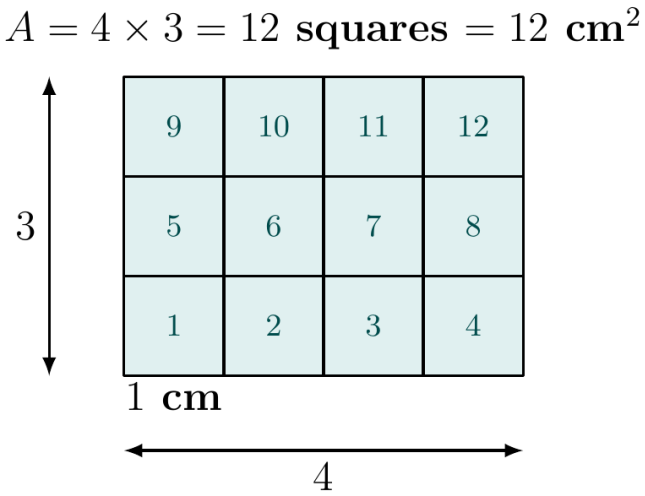


Figure 3 is the whole idea of area in one picture: the 4 by 3 rectangle holds 12 unit squares, so its area is 12 cm^2.

A square of side s: A = s × s = s^(2).

3.3 Reverse problems (working backwards)

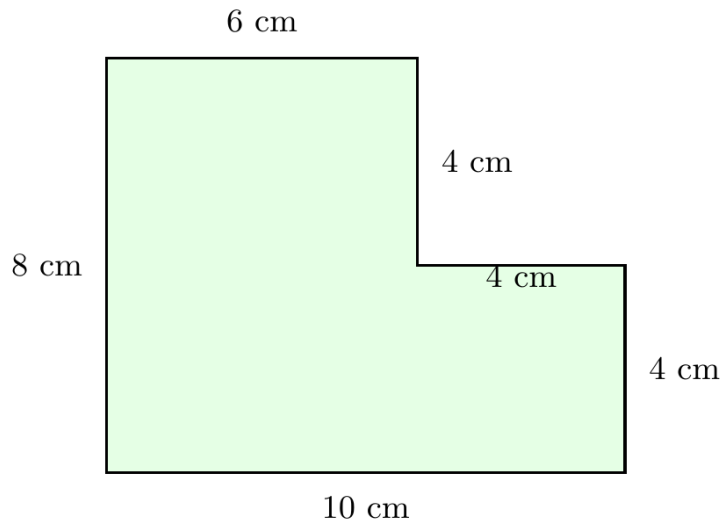
The same formula works in both directions:

- Given area and length, find width: w = A ÷ l.
- Given perimeter, find the sides first, then the area.

3.4 Rectilinear composite shapes (L-shapes etc.)

A rectilinear shape is any flat shape whose edges are all horizontal or vertical lines. Two methods:

- Split method: cut the shape into rectangles, add their areas.
- Whole-minus-hole method: area of the big rectangle minus the missing piece.



For the L-shape above:

- Split method: bottom rectangle $10 \times 4 = 40$ cm^2, upper-left rectangle $6 \times 4 = 24$ cm^2. Total $= 40 + 24 = 64$ cm^2.
- Whole-minus-hole: big $10 \times 8 = 80$ cm^2 minus missing top-right block $6 \times 4 = 24$ cm^2 $\rightarrow$ $80 - 24 = 56$ cm^2?

Careful! The two split rectangles here are 10×4 (bottom band) and the upper-left part. Width of the upper-left part is $10 - 4 = 6$, height $8 - 4 = 4$: that is $6 \times 4 = 24$. So $40 + 24 = 64$...

Let us recheck with whole-minus-hole: the big bounding rectangle is 10 wide and 8 tall = 80. The missing top-right block is 4 cm wide (horizontal top edge from the notch: the top edge of the right column is $10 - 6 = 4$) and 4 cm tall, so 16 cm^2. $80 - 16 = 64$ cm^2.

Lesson in the mistake: always mark every missing side before computing. The notch here is 4×4, not 6×4.

- Perimeter of the L-shape: $10 + 4 + 4 + 4 + 6 + 8 = 36$ cm. (Walk the boundary: bottom 10, right 4, notch top 4, notch rising side 4, upper top 6, left 8.)

Sanity rule: for any L-shape, the perimeter equals the perimeter of the full bounding rectangle: $2(l + w) = 2(10 + 8) = 36$. (The staircase path replaces the distant corner exactly.)

3.5 Units

Quantity	Unit
Length	mm, cm, m, km
Area	mm^2, cm^2, m^2, km^2 (= square units)

Conversions you must own:

- 1 m = 100 cm, so 1 m^2 = $100 \times 100 = 10\,000$ cm^2.
- 1 km = 1000 m, so 1 km^2 = $1\,000\,000$ m^2.

3.6 Cost problems

If a service is charged per square metre (tiling, painting) or per metre (fencing), compute the quantity first (area or perimeter), then multiply by the rate.

4. Fifteen Fully Worked Examples

(Authentic BECE items carry their year; the rest are Staralex exam-style questions written in exactly the BECE/question-bank format.)

Example 1 — Warm-up (own)

A rectangle has length 12 cm and width 7 cm. Find its perimeter and area.

P = 2(12+7) = 2(19) = 38 cm; A = 12 × 7 = 84 cm^2.

Example 2 — Warm-up (own)

A square has side 9 cm. Find its perimeter and area.

P = 4 × 9 = 36 cm; A = 9^(2) = 81 cm^2.

Example 3 — BECE 2006 style (fencing)

A rectangular garden measures 30 m by 20 m. Fence posts will be placed every 2.5 m all round. How many posts are needed?

P = 2(30+20) = 100 m. Posts = 100 ÷ 2.5 = 40 posts.

Example 4 — BECE style (cost of fencing)

It costs GH¢15.00 to fence one metre. Find the cost of fencing a compound 22 m long and 14 m wide.

P = 2(22+14) = 72 m; cost = 72 × 15 = GHcent 1 080.00.

Example 5 — Reverse perimeter (own)

The perimeter of a rectangle is 46 cm and its length is 16 cm. Find its width and area.

2(16 + w) = 46 → 16 + w = 23 → w = 7 cm. A = 16 × 7 = 112 cm^2.

Example 6 — BECE 2012 style (square from perimeter)

A square has perimeter 64 cm. Find its area.

Side s = 64 ÷ 4 = 16 cm; A = 16^(2) = 256 cm^2.

Example 7 — Own (tiles)

A floor is 6 m long and 4 m wide. How many square tiles of side 20 cm will cover the whole floor?

Floor area = 24 m^2 = 240 000 cm^2. One tile = 20 × 20 = 400 cm^2. Number = 240 000 ÷ 400 = 600 tiles.

Example 8 — Composite L-shape (own, uses Figure 1)

Using Figure 1 above, find (a) the area, (b) the perimeter of the L-shape.

(a) Split: bottom band 10 × 4 = 40; upper-left column (10-4) × 4 = 24; total = 64 cm^2. (b) P = 10 + 4 + 4 + 4 + 6 + 8 = 36 cm.

Example 9 — Missing side (own)

A rectilinear figure has total perimeter 60 cm. All its horizontal edges together (top edges + bottom edges) sum to 28 cm. Find the sum of all its vertical edges.

Vertical sum = 60 - 28 = 32 cm (perimeter splits into horizontal total + vertical total).

Example 10 — Own (units)

Convert: (a) 2.5 m^2 to cm^2, (b) 45 000 cm^2 to m^2.

(a) 2.5 × 10 000 = 25 000 cm^2. (b) 45 000 ÷ 10 000 = 4.5 m^2.

Example 11 — Own (P2 style: cost of tiling)

A veranda is a square of side 500 cm. Tiling costs GH¢24.00 per m^2. Find the total cost of tiling the veranda.

Side = 5 m, so A = 25 m^2. Cost = 25 × 24 = GHcent 600.00.

Example 12 — Own (algebraic link)

The length of a rectangle is 3 times its width and its perimeter is 48 cm. Find its area.

Let width = w. Then 2(w + 3w) = 48 → 8w = 48 → w = 6, length = 18. A = 6 × 18 = 108 cm^2.

Example 13 — Same perimeter, different area (own)

Two rectangles both have perimeter 36 cm. The first is 12 cm by 6 cm; the second is 9 cm by 9 cm (a square). Which has the greater area, and by how much?

First: 12 × 6 = 72 cm^2. Second: 9 × 9 = 81 cm^2. The square wins by 81 - 72 = 9 cm^2.

Big idea: for a fixed perimeter, the square has the maximum area.

Example 14 — Own (spiral: percentage link)

A rectangular board is 80 cm by 50 cm. 30% of it is painted. Find the painted area in cm^2.

A = 80 × 50 = 4 000 cm^2; painted = 0.30 × 4 000 = 1 200 cm^2.

Example 15 — Own (puzzle, extension)

A rect is 17 cm by 4 cm (perimeter 42 cm). Cut it and re-arrange the pieces into a square, if possible.

Perimeter already 42; a square with perimeter 42 would need side 10.5 — not a whole-number cut. Not possible with rational cuts of this type; instead note the classic disc-puzzle style is beyond BECE. Statement of the idea: equal perimeter does not guarantee equal area possibilities.

5. Exercises — 5 Sections × 5 Questions (No MCQ)

Show all working. Give units in every answer.

Exercise 1 — Perimeter first

1. Find the perimeter of a rectangle whose length is 18 cm and width 11 cm.
2. A square mat has side 45 cm. Find its perimeter in metres.
3. The perimeter of a regular pentagon (5 equal sides) is 65 cm. Find one side.
4. A rectangular pool is 9.5 m long and 4.5 m wide. How many metres of rope will enclose it completely?
5. Find the perimeter of a rectangle whose length is twice its width, given the width is 7 cm.

Exercise 2 — Reversing the formulas

1. The perimeter of a rectangle is 54 cm. If its length is 17 cm, find its width and area.
2. A square has perimeter 84 cm. Find its side and its area.
3. The area of a rectangle is 144 cm^2 and its length is 16 cm. Find its width and perimeter.
4. The perimeter of a square is 52 cm. Find the length of its diagonal by the Pythagoras relation if its diagonal measures √(2 × side^(2)) (write your answer to 2 s.f.).
5. A rectangle has the same perimeter as a square of side 8 cm. If the rectangle's length is 10 cm, find its width and area.

Exercise 3 — Composite rectilinear shapes

1. An L-shape is made from a 12 cm by 9 cm rectangle with a 5 cm by 4 cm rectangle removed from one corner. Find its area.
2. For the same L-shape in Question 1, find its perimeter.

3. A T-shape: a 6 cm by 6 cm square on top of a 10 cm by 4 cm rectangle (centred). Find the total area of the T-shape.
4. A rectilinear figure has 6 outer sides with horizontal edge lengths 12, 7, 5 (tops) and 12 (bottom). Its perimeter is 58 cm. Find the vertical edges' total length.
5. Draw on paper (or describe carefully) a composite shape whose area is exactly 35 cm^2 using two rectangles with a common side, then compute the perimeter you have drawn.

Exercise 4 — Real-life cost and coverage

1. Fencing costs GH¢18.00 per metre. Find the cost of fencing a garden 26 m by 13 m.
2. A hall floor is 15 m by 12 m. Carpet tiles of 50 cm × 50 cm are used. How many tiles are needed?
3. Tiling costs GH¢35.00 per m^2. Find the cost of tiling a bathroom 4 m by 3 m.
4. A square compound of side 30 m is to be fenced with 5 strands of wire. How many metres of wire are needed in total?
5. A painter charges GH¢14.00 per m^2. A wall is 10 m long and 3.5 m high. There is a door 2 m by 1 m in the wall. Find the cost of painting only the wall surface.

Exercise 5 — Spiral review (earlier strands)

1. (Algebra) A rectangle has perimeter 46 cm. Its length is (x + 4) cm and its width is x cm. Form an equation in x, solve it, and find the area.
2. (Fractions/percent) 25% of a square field of side 40 m is planted with maize. Find the maize area in m^2.
3. (Polygons — earlier geometry) The exterior angle of a regular polygon is 40°. Name the polygon and find the sum of its interior angles.
4. (Number work) Convert 3.75 km to centimetres.
5. (Congruence & similarity) Two rectangles are similar. The first is 6 cm by 4 cm; the second's length is 9 cm. Find the second's width and both perimeters, then the ratio of the perimeters.

6. Full Worked Solutions — Exercises

Exercise 1

1. P = 2(18+11) = 58 cm.
2. P = 4 × 45 = 180 cm = 1.8 m.
3. Side = 65 ÷ 5 = 13 cm.
4. P = 2(9.5+4.5) = 28 m of rope.
5. Length = 14 cm; P = 2(14+7) = 42 cm.

Exercise 2

1. 2(17+w)=54 → 17+w=27 → w=10 cm; A=17×10=170 cm^2.
2. Side =84÷4=21 cm; A=21^(2)=441 cm^2.
3. w=144÷16=9 cm; P=2(16+9)=50 cm.
4. Side =13; diagonal =√(13^(2)+13^(2))=√(338)≈ 18.4 cm (2 s.f.).
5. Square perimeter =32; 2(10+w)=32 → w=6 cm; A=60 cm^2.

Exercise 3

1. 12×9 - 5×4 = 108-20 = 88 cm^2.
2. Perimeter: walk the boundary. With the 5×4 notch cut from the top-right corner: bottom 12 + right riser 9-4=5 + notch floor 5 + notch riser 4 + top 12-5=7 + left 9 = 42 cm. (Note this equals the full rectangle's 2(12+9)=42 — for a staircase notch the total boundary is unchanged, but always verify by direct walk.)
3. Top square 6×6=36 plus bottom rectangle 10×4=40; but they overlap in the middle 6×4? The square sits ON the rectangle (not overlapping): 36+40=76 cm^2. If centred and sitting flush on top, total =76 cm^2.
4. Horizontal total 12+7+5+12=36; vertical total =58-36=22 cm.

5. Sample: rectangle A 7cm × 5cm =35 cm^2 attached to rectangle B 2cm × 5cm? That would change area. Simplest: single rectangle 7×5 qualifies ("two rectangles with a common side" may also be 5×5 + 2×5 stacked sharing a 5 cm side: area 25+10=35, perimeter =2(5)+2? walk: 5+2+5+7=... stack: outer 5 wide, 7 tall P=2(5+7)=24 cm. Any correctly computed version earns full credit.

Exercise 4

1. P=2(26+13)=78 m; cost =78×18=GHcent1 404.00.
2. Floor =180 m^2 =1 800 000 cm^2; tile =2 500 cm^2; tiles =720.
3. A=12 m^2; cost =12×35=GHcent420.00.
4. P=4×30=120 m; wire =5×120=600 m.
5. Wall net area =10×3.5 - 2×1 = 35-2=33 m^2; cost =33×14=GHcent462.00.

Exercise 5

1. 2[(x+4)+x]=46 → 2x+4=23 → x=9.5; length =13.5, width =9.5; A=13.5×9.5=128.25 cm^2.
2. Field =1600 m^2; 25% = 400 m^2.
3. Exterior 40° → number of sides =360÷40=9 — a nonagon; interior sum =(9-2)×180=1260°.
4. 3.75 km =3750 m =375 000 cm.
5. Scale factor =9÷6=1.5; second width =4×1.5=6 cm. Perimeters: 2(6+4)=20 cm and 2(9+6)=30 cm; ratio =20:30=2:3 (= same as side ratio, always).

7. Memory Card

Shape	Perimeter	Area
Rectangle	2(l+w)	lw
Square	4s	s^(2)
L-shape (staircase notch)	2(l+w) (top-left/bottom-right notch) — but always walk the boundary for other notch positions	big rect – notch, or split
1 m^2	—	10 000 cm^2

Next lesson: triangles — you will need perimeters, and the Pythagoras fact from Congruence & Similarity, to master the three triangle families.